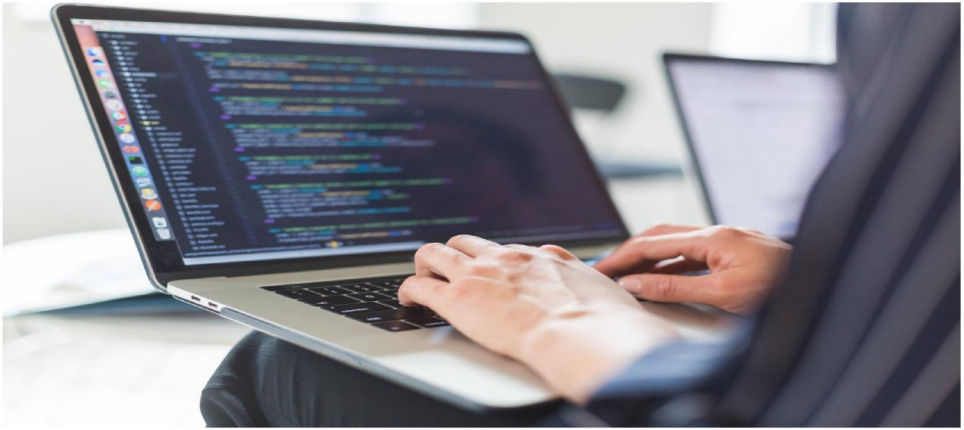


ETX1100/ETX5900 Business Statistics
Week 6: Regression Analysis & Correlation
Workshop Lead: Alyssa Hu

Reminders

Assignment due on 19 Jan 2026 at 11:55pm.

Consultation (dedicated student hours) – see Moodle for time/location.

Under *Own-Time* in Moodle, remember there is a Chatbot (All Day TA) and you can also practice unit-related questions using the Intelligent Quiz tab.

Outline of Topics

- Overview of Regression: Terminology and Concepts
- Examples and Working for Simple Linear Regression (SLR)
  - Scatterplot
  - Developing the Model
    - Slope, Intercept (and how to interpret)
    - Correlation (and how to interpret)
    - Prediction
    - Confidence Interval for the Slope
  - Evaluating the Model
    - R^2 (and how to interpret)
    - Hypothesis Testing of the Slope
- Getting Ready for Multiple Linear Regression (MLR)

Textbook references: Berenson et al *Basic Business Statistics 5 th ed*

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Overview of Regression: Terminology

Regression analysis is a study of relationships between variables We use sample data to develop a model of the relationship, which we can use to make predictions of the value of a numerical variable based on the value of one or more variables.

Some terminology:

- **Dependent variable:** the variable we wish to predict
- **Independent variable:** the variables used to make the prediction

Do not edit
How to change the design



For example, if a professor wants to predict student exam score based on how many hours a student studies. In this case, which is the dependent variable and which is the independent variable?

Overview of Regression: Terminology

Simple Linear Regression model:
Type of regression model that considers a **linear** function between a single dependent variable y and a single independent variable x.

Multiple Linear Regression model:
Type of regression model that considers a **linear** function between a single dependent variable y and multiple independent variables

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Continuing our example: If a professor wants to predict student exam score based on how many hours a student studies, is this appropriate for simple linear regression or multiple linear regression?

Note!

Today, we will mainly focus on simple linear regression (SLR).

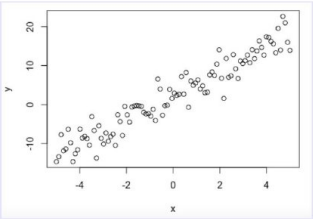
Next week, we extend what we've learned into multiple linear regression (MLR). We'll also learn about time series data.

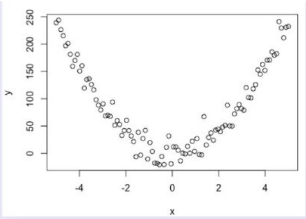
Overview of SLR Concepts

Good to do before modeling: **Create a scatter plot of Y vs X**

It is always advisable to graph the two variables to see whether it is reasonable to use a linear model. If we see a quadratic or exponential relationship, a linear model doesn't seem appropriate.

Plot A





Plot B

On the next slide, we will have a Slido question about which plot (A, B, both, neither) is appropriate to proceed with SLR. Unfortunately, I can't embed images in the Slido, so take a look at these plots and

Do not edit
How to change the design



Which is the correct statement?

Overview of SLR Concepts

Quantify (specify & estimate) the relationship by fitting a model:
Y as a function of X

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

- Y_i: dependent variable (also called response variable) for observation *i*
- X_i: independent variable (also called explanatory variable) for observation *i*
- β_0: y-intercept for the population (value of Y when X = 0)
- β_1: slope for the population (calculated as $\frac{\Delta Y}{\Delta X}$)
- ε_i: random error for observation *i*

Overview of SLR Concepts

Quantify (specify & estimate) the relationship by fitting a model:

Hmm… we usually don't have information from the population. We'll need to estimate from our sample – be careful with notation.

$$\hat{Y}_i = b_0 + b_1 X_i$$

- $\hat{Y}_i$: predicted value of Y for observation *i*
- X_i: value of X for observation *i*
- b_0: sample y-intercept (value of $\hat{Y}$ when X = 0)
- b_1: sample slope (change in **predicted** Y, **on average**, when the value of X **increases by one unit**)
  - If b_1 is positive, an increase in X is accompanied by an increase in Y
  - If b_1 is negative, an increase in X is accompanied by a decrease in Y

Overview of SLR Concepts

$$\hat{Y}_i = b_0 + b_1 X_i$$

We have X_i from the data. How exactly do we get these b_0, b_1?

We need to have some criteria for the line we're modelling. (Drawing a random line on a page isn't helpful.) So, we will use an approach called **the least squares method,** which Excel will help us implement.

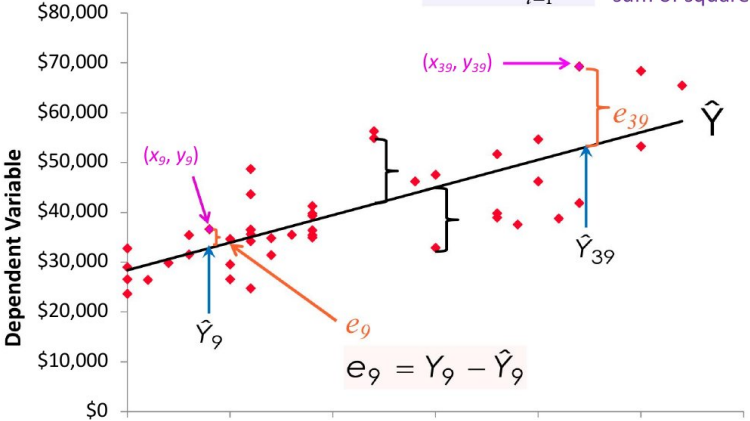
The idea is that we want a line that minimizes the sum of squares ("least squares") of the *errors.*

Overview of SLR Concepts

$$e_i = Y_i - \hat{Y}_i$$

$$SSE = \sum_{i=1}^{n} e_i^2$$

Collective errors are summarised by the sum of squared error (SSE)



Outline of Topics

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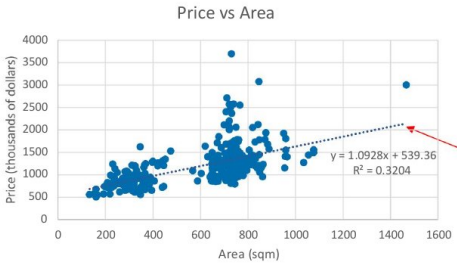
Task 1: Real Estate Data (Guided Example)

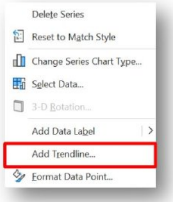
Using the **Workshop 6 Workshop Data** titled **RealEstate**:

- Draw a scatter plot of property price (Y, in thousands of dollars) against area (X, in sqm). Add a trendline to the plot.
- Generate a simple linear regression output in Microsoft Excel with price as the dependent variable and area as the independent variable.
  - Report the intercept and slope coefficient. Interpret them, explaining what it means in the context of property prices.
  - Report the correlation coefficient. Interpret what it tells you about the strength and direction of the relationship between area and price.
- Using the regression model, compare the predicted prices of two properties, one with an area of 600 sqm and another with 900 sqm. State how much higher the predicted price is for the larger property. Explain why.

Task 1: Real Estate Data (Guided Example)

Draw a scatter plot of Property Price (Y) against Area (X, in sq. m). Add a trendline to the plot.



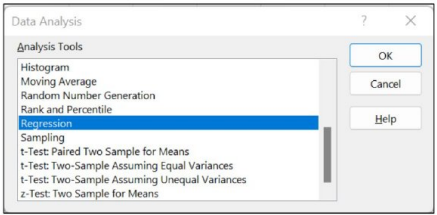


Positive slope!

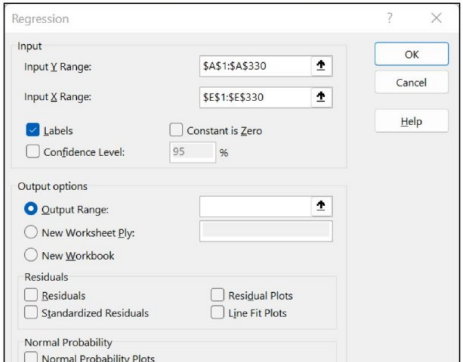
Task 1: Real Estate Data (Guided Example)

Generate a simple linear regression output in Microsoft Excel with Price as the dependent variable and Area as the independent variable.

Data > Data Analysis > Regression



Y = Price X = Area



Tick Labels as the column names were included, and specify a cell for the Output Range.

Task 1: Real Estate Data (Guided Example)

Generate a simple linear regression output in Microsoft Excel with Price as the dependent variable and Area as the independent variable.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.565997549					
R Square	0.320353226					
Adjusted R Square	0.318274795					
Standard Error	345.8900663					
Observations	329					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	1	18440376.06	18440376.06	154.132277	2.9314E-29	
Residual	327	39122259.72	119639.938			
Total	328	57562635.78				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	539.3571773	57.70092575	9.34746142	1.4529E-18	425.8453136	652.8690411

Task 1: Real Estate Data (Guided Example)

Report the intercept and slope coefficient. Interpret them, explaining what it means in the context of property prices.

SUMMARY OUTPUT							
Regression Statistics							
Multiple R	0.565997549						
R Square	0.320353226						
Adjusted R Square	0.318274795						
Standard Error	345.8900663						
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Total	328	57562635.78					
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	
Intercept b_0	539.3571773	57.70092575	9.34746142	1.4529E-18	425.8453136	652.8690411	
Area b_1	1.093	0.088060	12.41	2.93E-29	0.919	1.266	

Model: $\widehat{Price}_i = 539.357 + 1.093 Area_i$

Task 1: Real Estate Data (Guided Example)

Report the intercept and slope coefficient. Interpret them, explaining what it means in the context of property prices.

Intercept:
The intercept of 539.357 tells us that the expected price of a property with Area = 0 sqm is $539,357.
Does this make sense? **Does the intercept have a meaningful interpretation?**

Slope:
The slope of 1.093 indicates that an additional square meter in land area corresponds to an increase of $1,093 in property price, on average.

What is correlation?

Correlation is a measure of the relative strength of the **linear** relationship between two variables, taking values between -1 and 1.

Usually, we will not see *perfect* negative linear correlation (-1) or *perfect* positive linear correlation (+1).

- Strength:
  - If correlation is close to -1 or +1, it's a relatively strong correlation.
  - If correlation is close to 0, it is relatively weak correlation.
  - Otherwise, it is relatively moderate.
- Direction: Positive (+), negative (-)
- ρ (notation for population), r (notation for sample)

Task 1: Real Estate Data (Guided Example)

Report the correlation coefficient. Interpret what it tells you about the strength and direction of the relationship between area and price.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.565997549
R Square	0.320353226
Adjusted R Square	0.318274795
Standard Error	345.8900663
Observations	329

For simple linear regression, this is the absolute value of the correlation coefficient, also written as $|r|$.

You must check the sign of the slope to determine r itself!

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	18440376.06	18440376.06	154.132277	2.9314E-29
Residual	327	39122259.72	119639.938		
Total	328	57562635.78			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	539.3571773	57.70092575	9.34746142	1.4529E-18	425.8453136	652.8690411

Task 1: Real Estate Data (Guided Example)

Report the correlation coefficient. Interpret what it tells you about the strength and direction of the relationship between area and price.

$r = 0.566$

(We use *r* because this is calculated from our *sample* data.)

We consider this to be moderate and positive correlation between Area (sqm) and Price (in thousands of dollars).

Again, the direction of the correlation and sign of the slope must be the same. As the area increases, the property price also tends to increase.

Task 1: Real Estate Data (Guided Example)

Using the regression model, compare the predicted prices of two properties, one with an area of 600 sq. m and another with 900 sq. m. State how much higher the predicted price is for the larger property. Explain why.

$$\widehat{Price} = 539.357 + 1.093 Area$$

Note: You should use the exact numbers from the regression output when it is available.

$\widehat{Price} = 539.357 + 1.093 * 900$ = \$1,522,854
$\widehat{Price} = 539.357 + 1.093 * 600$ = \$1,195,021

Difference = \$327,832

Another way to calculate the diffference: slope * (900-600)

Task 1: Real Estate Data (Guided Example)

Report and interpret the confidence interval for the slope .

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.565997549					
R Square	0.320353226					
Adjusted R Square	0.318274795					
Standard Error	345.8900663					
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ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
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Intercept	539.3571773	57.70092575	9.34746142	1.4529E-18	425.8453136	652.8690411

CI for b_1

Task 1: Real Estate Data (Guided Example)

Report and interpret the confidence interval for the slope.

CI in terms of thousands of dollars: ($920, $1266)

We are 95% confident that the price increase for an additional square meter of land is between $920 and $1,266 on average.

Task 2: Marketing (Your Turn)

Using the **Workshop 6 Workshop Data** titled **Marketing**:

- What do you expect the relationship to be between spend and income?
- Draw a scatter plot of spend (Y, in dollars) against income (X, in dollars). Add a trendline to the plot. Comment on how this compares to your expectation in Question 1.
- Generate a simple linear regression output in Microsoft Excel with spend as the dependent variable and income as the independent variable.
  - Report the intercept and slope coefficient. Interpret in context.
  - Report the correlation coefficient. Interpret in context what it tells you about the strength and direction of the relationship.
- What is the estimated or predicted spend when income is $50000?
- Interpret the confidence interval for the slope.

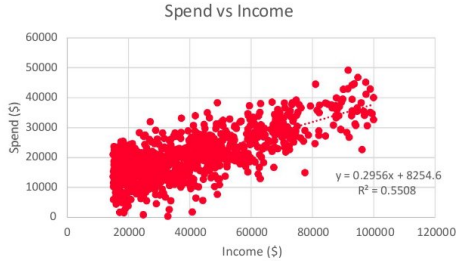
Task 2: Marketing (Your Turn)

What do you expect the relationship to be between spend and income?

I would generally expect Spend and Income to be positively correlated, since a higher income would tend to allow the consumer to spend more on non-essential items.

Task 2: Marketing (Your Turn)

Draw a scatter plot of spend (Y, in dollars) against income (X, in dollars). Add a trendline to the plot. Comment on how this compares to your expectation in Question 1.



Positive slope - as the Income increases, amount spent by the consumer also increases.

Task 2: Marketing (Your Turn)

Generate a simple linear regression output in Microsoft Excel with spend as the dependent variable and income as the independent variable.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.742137939
R Square	0.550768721
Adjusted R Square	0.55020928
Standard Error	5594.82837
Observations	805

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	30816858388	30816858388	984.4979719	1.1097E-141
Residual	803	25135589906	31302104.49		
Total	804	55952448293			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	8254.575214	445.0118351	18.5491139	3.37026E-64	7381.051414	9128.099015
Income	0.295614568	0.009421465	31.37671066	1.1097E-141	0.277120962	0.314108174

Task 2: Marketing (Your Turn)

Report the intercept and slope coefficient. Interpret in context.

$$\hat{y} = b_0 + b_1 X$$

$$\widehat{Spend} = 8254.575 + 0.296 Income$$

Intercept:
The intercept of 8254.575 tells us that the expected spending of someone with $0 in income is approximately $8255 dollars.

Slope:
The slope of 0.296 indicates that an additional thousand dollars in income corresponds to an increase of approximately $296 in spending, on average.

Task 2: Marketing (Your Turn)

Report the correlation coefficient. Interpret in context what it tells you about the strength and direction of the relationship.

$r = 0.742$

We consider this to be moderate/strong and positive correlation between income ($) and spending ($).

Again, the direction of the correlation and sign of the slope are the same. As income increases, spending increases.

Task 2: Marketing (Your Turn)

What is the estimated or predicted spend when income is $50000?

$$\widehat{Spend} = 8254.575 + 0.296Income$$

Plug in $50000 for *X*.

The estimated spend is $23035 when income is $50000.

Task 2: Marketing (Your Turn)

Interpret the confidence interval for the slope.

CI: (.277, .314)

We are 95% confident that the spend increase for an additional thousand dollars in income is between $277 and $314 on average.

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    - Confidence Interval for the Slope
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    - R^2 (and how to interpret)
    - Hypothesis Testing of the Slope
- Getting Ready for Multiple Linear Regression (MLR)

Textbook references: Berenson et al *Basic Business Statistics 5 th ed*

Task 1: Real Estate Data (Guided Example)

Using the **Workshop 6 Workshop Data** titled **RealEstate**:

- Interpret the coefficient of determination R^2, and explain how well the model explains variation in property prices.
- Test whether the regression slope is statistically significant.

What is the coefficient of determination?

$$Y_i = \underbrace{\beta_0 + \beta_1 X_i}_{\text{explainable}} + \underbrace{\varepsilon_i}_{\text{unexplained}}$$

The **coefficient of determination** is the portion of the total variation in the dependent variable (Y) that is ***explained by*** the linear relationship with the independent variable (X).

R^2 is a measure of 'Goodness of Fit', often expressed in %. If we have a good model, the unexplained component is small.

$0 \leq R^2 \leq 1$

$R^2 = 0$ (or 0%) means no linear relationship between X and Y

$R^2 = 1$ (or 100%) means deterministic linear relationship between X and Y

What is the coefficient of determination?

Total Sum of Squares = **Regression Sum of Squares** + **Error Sum of Squares**

$$SST = \sum (Y_i - \bar{Y})^2$$

Measures the **TOTAL** variation of the Y_i values around their sample mean $\bar{Y}$

$$SSR = \sum (\hat{Y}_i - \bar{Y})^2$$

- Part of the total variation that can be explained by the regression.
- Measures the **EXPLAINED** variation attributable to the relationship between X and Y.
- This measures the improvement of $\hat{Y}_i$ over the $\bar{Y}$ as the predictor of Y

$$SSE = \sum (Y_i - \hat{Y}_i)^2$$

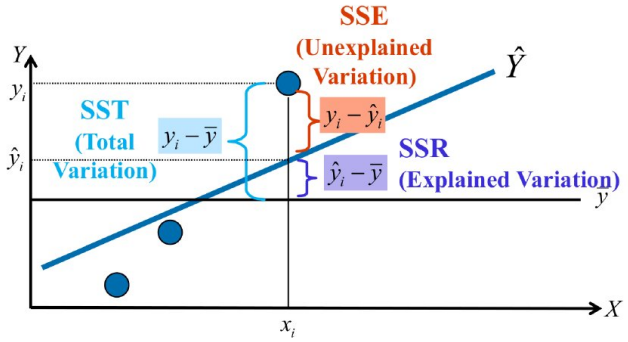
- Part of the total variation that is unexplained by the model.
- Measures the **UNEXPLAINED** Variation attributable to factors other than the relationship between X and Y.

SST = SSR + SSE

A model fits the data well if **SSR** is a "large" proportion of **SST**

$$R^2 = \frac{SSR}{} = \frac{\text{Regression Sum of Squares}}{}$$

What is the coefficient of determination?



SST = **SSR** + **SSE**

A model fits the data well if **SSR** is a "large" proportion of **SST**

$$R^2 = \frac{SSR}{} = \frac{\text{Regression Sum of Squares}}{}$$

Task 1: Real Estate Data (Guided Example)

Interpret the coefficient of determination R^2, and explain how well the model explains variation in property prices.

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.565997549
R Square	0.320353226
Adjusted R Square	0.318274795
Standard Error	345.8900663
Observations	329

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
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Total	328	57562635.78			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	539.3571773	57.70092575	9.34746142	1.4529E-18	425.8453136	652.8690411

Task 1: Real Estate Data (Guided Example)

Interpret the coefficient of determination R^2, and explain how well the model explains variation in property prices.

The R^2 value is 32% for the simple linear model. About 32% of the variation in property prices can be explained by the model.

About 68% of the variation in property prices cannot be explained by the model.

Remember hypothesis testing?

The least squares method will produce a regression line regardless of whether there is actually a linear relationship between X and Y, so we need to use hypothesis testing on the regression slope coefficient.

Once again, we will need to use our 5 steps (see next 5 slides).

HT of Regression Slope – Step 1

Our null hypothesis says that there is NO linear relationship between *X* and *Y*.

Our alternative hypothesis is where we make claims about a linear relationship.

H_0: $\beta_1 = 0$
H_1: $\beta_1 \neq 0$
Two-tailed test. There is a linear relationship between X & Y.

H_0: $\beta_1 = 0$ (***or:*** H_0: $\beta_1 \geq 0$)
H_1: $\beta_1 < 0$
Left-tailed test. There is a negative linear relationship between X & Y.
X has a negative effect on Y

H_0: $\beta_1 = 0$ (***or:*** H_0: $\beta_1 \leq 0$)
H_1: $\beta_1 > 0$
Right-tailed test. There is a positive linear relationship between X & Y.
X has a positive effect on Y

HT of Regression Slope – Steps 2 and 3

Step 2. The test statistic is $t = \frac{(b_1 - \beta_{1,H_0})}{SE(b_1)}$. We can use Excel to get this.

β_{1,H_0} = the hypothesized null value of the population slope (which is 0)

b_1 = the sample estimate of β_1

$SE(b_1)$ = the standard error of b_1

The test statistic follows a t-distribution with $n - 2$ degrees of freedom.

Step 3. To find the p-value, we will also use Excel ☺

Warning: Excel by default will give a p-value for the two-tailed test. If you are doing a one-tailed test, you MUST divide by 2 to get p-value

HT of Regression Slope – Steps 4 and 5

Step 4. Same decision rule.

Reject H_0 if *p-value* < α
Fail to reject H_0 otherwise

Step 5. Conclusion

If p-value < α, we **CAN** reject H_0 There is sufficient evidence at the (α)% level of significance that there is a significant linear relationship between X and Y in the population based on the sample data.	If p-value > α, we **CANNOT** reject H_0 There is insufficient evidence at the (α)% level of significance that there is a significant linear relationship between X and Y in the population based on the sample data.

Task 1: Real Estate Data (Guided Example)

Test if there is a linear relationship:

Default approach adopted by statistical software packages is to apply two-sided tests.

Hypotheses:
$H_0: \beta_1 = 0$
$H_1: \beta_1 \neq 0$

***Level of significance**:α=0.05*
***t-statistic** = 12.415*
p-value = 2.9314E-29 ≈ 0
(from the Excel output)

SUMMARY OUTPUT

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Multiple R	0.565997549
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Adjusted R Square	0.318274795
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ANOVA

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Residual	327	39122259.72	119639.938		
Total	328	57562635.78			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	539.3571773	57.70092575	9.34746142	1.4529E-18	425.8453136	652.8690411
Area	1.092773852	0.088020432	12.41500209	2.9314E-29	0.919616087	1.265931617

***Decision Rule**:*
Reject H_0 if *p-value* < α
Since *p-value* = 0.0000 < 0.05, reject H_0

***Conclusion**:*
There is sufficient evidence at the 5% level of significance to conclude

Task 1: Real Estate Data (Guided Example)

Test if there is a positive linear relationship:

Default approach adopted by statistical software packages is to apply two-sided tests.

Hypotheses:

$H_0: \beta_1 = 0$

$H_1: \beta_1 > 0$

***Level of significance**:α=0.05*

***t-statistic** = 12.415*

***p-value** = **2.9314E-29/2** ≈ 0*

(from the Excel output)

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.565997549					
R Square	0.320353226					
Adjusted R Square	0.318274795					
Standard Error	345.8900663					
Observations	329					
ANOVA						
	df	*SS*	*MS*	*F*	*Significance F*	
Regression	1	18440376.06	18440376.06	154.132277	2.9314E-29	
Residual	327	39122259.72	119639.938			
Total	328	57562635.78				
	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	539.3571773	57.70092575	9.34746142	1.4529E-18	425.8453136	652.8690411
Area	1.092773852	0.088020432	12.41500209	2.9314E-29	0.919616087	1.265931617

***Decision Rule**:*

Reject H_0 if *p-value* < α

Since *p-value* = 0.0000 < 0.05, reject H_0

***Conclusion**:*

There is sufficient evidence at the 5% level of significance to conclude there is a

Task 2: Marketing (Your Turn)

Using the **Workshop 6 Workshop Data** titled **Marketing**:

- Interpret the coefficient of determination R^2, and explain how well the model explains variation in spend.
- Using a 5% level of significance, conduct a hypothesis test of the slope to determine if a **positive linear relationship** exists between spend and income.

Reminder: Excel Output

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.742137939
R Square	0.550768721
Adjusted R Square	0.55020928
Standard Error	5594.82837
Observations	805

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	30816858388	30816858388	984.4979719	1.1097E-141
Residual	803	25135589906	31302104.49		
Total	804	55952448293			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	8254.575214	445.0118351	18.5491139	3.37026E-64	7381.051414	9128.099015
Income	0.295614568	0.009421465	31.37671066	1.1097E-141	0.277120962	0.314108174

Task 2: Marketing (Your Turn)

Interpret the coefficient of determination R^2, and explain how well the model explains variation in spend.

R^2 = 55.08%

Approximately 55.08% of the **variation** in Spend is explained by the model.

The remaining 44.92% of **variation** in Spend is left unexplained by the model.

Task 2: Marketing (Your Turn)

Using a 5% level of significance, conduct a hypothesis test of the slope to determine if a **positive linear relationship** exists between spend and income.

Step 1:

$H_0: \beta_1 = 0$

$H_1: \beta_1 > 0$

Step 2:

α = 0.05

t-statistic = 31.377

Step 3: p-value

Step 4:

Reject H_0 if p-value < α

Since 1.1097E-141/2 < 0.05,

we can reject H_0

Step 5:

There is sufficient evidence at the 5% level of significance to conclude that there is a positive linear relationship

Outline of Topics

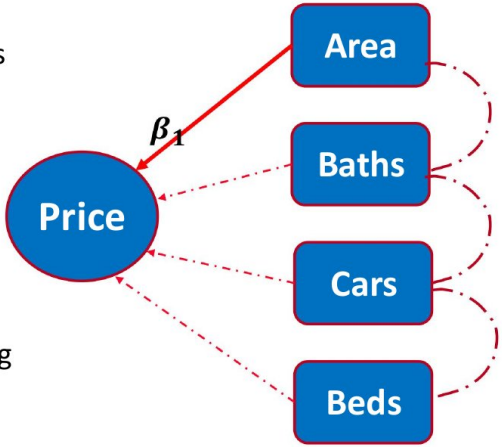
- Overview of Regression: Terminology and Concepts
- Examples and Working for Simple Linear Regression (SLR)
  - Scatterplot
  - Developing the Model
    - Slope, Intercept (and how to interpret)
    - Correlation (and how to interpret)
    - Prediction
    - Confidence Interval for the Slope
  - Evaluating the Model
    - R^2 (and how to interpret)
    - Hypothesis Testing of the Slope
- Getting Ready for Multiple Linear Regression (MLR)

Simple Linear Regression Pitfalls

So far, we've only included a SINGLE independent variable.

Omitting other relevant variables from the regression model may result in biased estimates of the regression coefficient (omitted variable bias).

e.g. Larger properties (in land area) tend to have larger dwelling size and therefore, more beds, baths and cars.



Simple Linear Regression Pitfalls

So we *could* estimate several simple linear regressions:

	Intercept	Slope
Area	539.357	1.093
Beds	261.501	264.855
Baths	661.001	288.180
Cars	845.000	200.464

These results seemingly give you area, beds, baths and cars premiums (i.e. how they increase the price).

But they are misleading because of the interconnected relations between variables!

Multiple Linear Regression (MLR)

The dependent variable Y may depend on several independent variables… (not just one!)

$$Y_i = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + e_i$$

Independent variables $X_1, X_2, \ldots, X_k$ are related to Y. But some of the independent variables may also be related.

If you hold $X_2, \ldots, X_k$ constant, how would X_1 affect Y?

We can generate a multiple regression output in Excel. Next time, we will select a number of columns for the independent variables in the 'Input X Range'

Multiple Linear Regression (MLR)

$$\hat{Y}_i = b_0 + b_1 X_1 + b_2 X_2 + \cdots + b_k X_k$$

b_0: The predicted value of Y when all X variables are zero
Not meaningful when the range value of any X variables does not include 0

b_1: The change in predicted Y when X_1 value increases by one unit, holding other X variables fixed.

b_k: The change in predicted Y when X_k value increases by one unit, holding other X variables fixed.

Important: The slope estimates in MLR are **marginal effects**, i.e. the effect of a single X input holding other X variables fixed.
(In SLR, the slope tells you the *total effect* of X on Y because there